

Bangyu Li

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Online portfolio: bangyuli.vercel.app

EDUCATION

Newcastle University

Newcastle upon Tyne, U.K.

MSc Computer Game Engineering

Expected: Sep 2025

Relevant Coursework:

- **Advanced Graphics for Games:** Focused on real-time rendering, lighting, and polygon mesh models, providing a deep understanding of graphics techniques used in modern game engines.
- **Advanced Game Technologies:** Covered collision detection, physics, and AI implementation, providing deep insight into game engine architecture and development.

University at Buffalo, SUNY

Buffalo, NY, U.S.

BSc Computer Science

Jan 2018 – Aug 2021

Relevant Coursework & Projects:

- **CSE442 - Software Engineering:** Developed a full-stack web application as part of a group project using React and Node.js, *Project Frame*, a Wiki-style community platform for fighting game players.
- **CSE474 – Intro. to Machine Learning:** Built a COMPAS-style risk assessment system using Naïve Bayes and predictive parity. Collaborated on data collection and model evaluation in a team setting.

PROFESSIONAL EXPERIENCE

Breaker Games, Shanghai

Oct 2023 – Apr 2024

UE5 Technical Designer

- Designed and Implemented DLC content for Project Zero 2: Apocalypse Party using Blueprints, including weapons and character talents. The DLC increased player retention by 24% on Steam and received mostly positive reviews.
- Collaborated closely with external translation team and led the localization of PZ2: Apocalypse Party.

Treelab Software, Shanghai

Jun 2022 – Apr 2023

Golang Back-end Engineer

- Developed the workflow execution service using a Temporal + RabbitMQ architecture, improving reliability for specific business scenarios by over 50%.
- Maintained infrastructure functions under a Docker microservice architecture, including logging, l18n (Internationalization) and data synchronization, ensuring stable infrastructure performance.
- Built CI/CD pipelines for automated testing and deployment, significantly accelerating project delivery.

Lilith Games, Shanghai

Jun 2021 – Jun 2022

Technical Designer

- Collaborated with cross-functional teams and oversaw acceptance testing for physics, terrain, and matchmaking systems in Avatar game engine.
- Developed plugin tools that provided versatile technical solutions for the game design and art teams, significantly improving the production pipeline efficiency.
- Designed and implemented multiple gameplay features for Project Party, including collecting mechanics and inventory systems, contributing to a more engaging player experience.
- Reconstructed the Cutscene Editor and designed the quest system for a narrative-driven game framework (Revolver Editor), enhancing the storytelling mechanics and the usability of Avatar engine.

PROJECTS

Light CatRun – [TapTap](#) | Demo Video: [Youtube](#)

An adorable strategic puzzle adventure. Entered the 2024 TapTap Spotlight GameJam, nominated for 'Best Game', 'Best Theme & Narrative', and 'Best Visual & Audio', and proudly awarded 'Best Visual & Audio'.

Role: UI programmer, Gameplay programmer. Designed and implemented core gameplay mechanics, including UI and interactable objects.

Fantasy RPG Demo – [Github](#) | Demo Video: [Bilibili](#)

A MMORPG demo showcasing my expertise in designing and implementing complex gameplay systems.

- **Networking:** Implemented a C# network messaging framework using Protobuf for real-time, event-driven state synchronization of character movement, ensuring smooth and reliable multiplayer interactions.
- **Game Systems:** Developed core gameplay systems, including inventory, shop, friend list, team-up, and multi-channel chat to enrich multiplayer gameplay experience.
- **User Interface:** Built an intuitive UI framework with UGUI for managing menus, item displays, and inventory interactions, improving the overall user experience and interface responsiveness.

Tic-Tac-Toe – [Github](#)

Implemented the Tic-Tac-Toe game with AI opponents of varying difficulty, utilizing algorithms like minimax for unbeatable difficulty level.

Grappling Hook Demo – [Github](#)

Recreated the grappling hook mechanics from *Sekiro: Shadows Die Twice* using Unity. Implemented dynamic rope physics and hook detection to create realistic grappling behavior.

SKILLS:

Programming languages: Proficient in *Python / Go / C# / C++ / Lua*

Unity Engine: Proficient in *UGUI / Unity plugins / prototyping*
Experienced with *Animator / Rigging / Shader*

Unreal 5 Engine: Proficient in *Blueprint scripting / prototyping*
Experienced with *Able Ability System / Behavior tree*

Database systems: Proficient in *MySQL / Oracle Database / DBeaver*
Experienced with *MongoDB / SQL Server*

Version control tools: Proficient in *Git / Perforce*